Topic: Time-Frequency Uncertainty Limitations in Function Transforms

Research Question:

To what extent does the effectiveness of STFT versus CWT in achieving precise time-frequency resolution depend on whether the signal is stationary or time-varying in frequency?

Word Count: 3985 Words

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1 Introduction and Motivation

Whenever a signal is interpreted, there are two perspectives (illustrated in Figure 1): the frequency domain, with each frequency that makes up the signal, and the time domain, or the signal at each moment in time. Based on the scenario, such as in signal processing, either of these domains may be required. Ideally, however, we would want perfect time resolution and perfect frequency resolution simultaneously. As per the Gabor-Heisenberg uncertainty principle, this is not possible.[1]

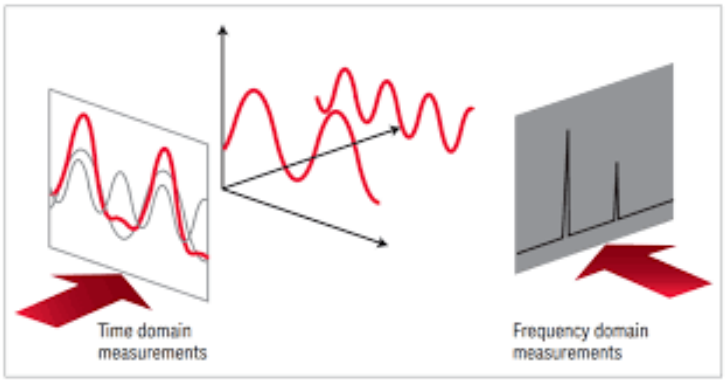


Figure 1: Frequency Domain vs. Time Domain

When you listen to a musical note that warbles in pitch, or when a radar sends out a chirp signal whose frequency increases over time, the sound's frequency content is constantly changing. Traditional Fourier analysis cannot tell us *when* each frequency occurs because it only gives the overall frequency content. Two mathematical tools solve this problem: the Short-Time Fourier Transform (STFT)

[1]Dennis Gabor. "Theory of Communication," Nov. 1944, jcsphysics.net/lit/gabor1946.pdf. Accessed 1 Feb. 2026.

and the Continuous Wavelet Transform (CWT).[2] Both are limited by the time-frequency uncertainty principle discussed earlier. This raises a natural question: **which transform gives better results for different types of signals?**

Research Question: To what extent does the effectiveness of STFT versus CWT in achieving precise time-frequency resolution depend on whether the signal is stationary or time-varying in frequency?

2 Methodology

To quantitatively measure how well each transform localizes energy in time and frequency, we analyze the basis functions each uses. STFT fundamentally applies Fourier with a window $h(t)$ that slides across the signal, allowing for time localization. Meanwhile, CWT has wavelets $\psi_{a,b}(t)$ that are both scaled and shifted, also providing partial time and frequency resolution. $U[g] = \sigma_t[g] \cdot \sigma_\omega[g]$ is the uncertainty product we can use to represent the time-frequency concentration of any basis function, allowing for direct comparison between these transforms.

Section 3 develops variance measures for finite-energy functions. Sections 4 and 5 derive $U[h] = 1/2$ for the Gaussian STFT window and $U[\psi_{a,b}] = 1/2$ for the Morlet wavelet, establishing both transforms achieve theoretical optimality. Section 6 describes two test signals that have been optimally selected: a Gaussian pulse $f_1(t) = e^{-t^2/2}$, where the STFT would perform well due to its stationary nature, and linear chirp $f_2(t) = \cos(t^2)$ with time-varying frequency such that CWT will likely outperform. Section 7 validates analytical results computationally

[2] Andrew Nicoll. "The Wavelet Transform for Beginners," July 2020, youtu . be / kuuUaqAjeoA. Accessed 1 Feb. 2026.

and compares representation concentration U_{rep} for each transform-signal combination, directly answering which achieves better resolution for stationary versus time-varying signals.

3 Mathematical Framework for Any Finite-Energy Function

3.1 What is a Finite-Energy Function?

We consider functions $g : \mathbb{R} \to \mathbb{R}$ that have finite total energy[3], meaning

$$E_g = \int_{-\infty}^{\infty} |g(t)|^2 dt$$

is a finite positive number. This condition ensures that integrals converge. Both test signals (Gaussian pulse and chirp) and basis functions (STFT windows and CWT wavelets) satisfy this.

3.2 Normalized Energy Density

For any function in the L^2 space, we can hence normalize the energy density:

[3]Pu Sun. "Comparison of STFT and Wavelet Transform in Time-Frequency Analysis," Dec. 2014, www.diva-portal.org/smash/get/diva2:793176/FULLTEXT01.pdf. Accessed 1 Feb. 2026.

$$\rho_g(t) = \frac{|g(t)|^2}{E_g}$$

The numerator represents the instantaneous energy at time t and the denominator is the total energy, making $\rho_g(t)$ energy density.

By normalizing, $\rho_g(t)$ satisfies the properties of a probability density function: $\rho_g(t) \geq 0$ and $\int_{-\infty}^{\infty} \rho_g(t)dt = \frac{1}{E_g}\int_{-\infty}^{\infty} |g(t)|^2 dt = 1$. This enables applying statistical measures of spread to characterize energy distribution over time.

3.3 Time-Domain Mean and Variance

As such, we have the mean time of a function:

$$\mu_t[g] = \int_{-\infty}^{\infty} t \cdot \rho_g(t)dt = \frac{1}{E_g}\int_{-\infty}^{\infty} t|g(t)|^2 dt$$

And the time variance, or the energy spread in the time domain:

$$\sigma_t^2[g] = \int_{-\infty}^{\infty} (t - \mu_t[g])^2 \rho_g(t)dt = \frac{1}{E_g}\int_{-\infty}^{\infty} (t - \mu_t[g])^2 |g(t)|^2 dt$$

The time standard deviation, $\sigma_t[g] = \sqrt{\sigma_t^2[g]}$, quantifies time localization, where a smaller $\sigma_t[g]$ indicates energy concentrated in a narrow time window.

3.4 Fourier Transform: Decomposing into Frequencies

To analyze frequency content, we use the Fourier transform:

$$G(\omega) = \int_{-\infty}^{\infty} g(t)\, e^{-i\omega t}\, dt$$

This integral measures how much of each frequenc3y ω is present in $g(t)$. For a Gaussian $a(t) = e^{-t^2/2}$, the transform is another Gaussian $A(\omega) = \sqrt{2\pi}\, e^{-\omega^2/2}$, showing an example of a function concentrated in time that is also concentrated in frequency.

The Fourier transform provides an energy distribution across frequencies by converting a time-domain function $g(t)$ into its frequency-domain representation $G(\omega)$, allowing use of the same variance concepts.

3.5 Frequency-Domain Energy and Variance

Parseval's theorem[4] states that total energy is preserved under the Fourier transform:

$$E_g = \int_{-\infty}^{\infty} |g(t)|^2 dt = \frac{1}{2\pi} \int_{-\infty}^{\infty} |G(\omega)|^2 d\omega.$$

The factor $\frac{1}{2\pi}$ arises from using angular frequency ω (rad/s) rather than ordinary frequency f (Hz). Thus $\frac{|G(\omega)|^2}{2\pi}$ is the energy density in frequency. Normalizing, we define the mean frequency and frequency variance, such that $\sigma_\omega[g]$ represents the typical frequency width.

$$\mu_\omega[g] = \int_{-\infty}^{\infty} \omega\, \tilde{\rho}_g(\omega)\, d\omega$$

$$\sigma_\omega^2[g] = \int_{-\infty}^{\infty} (\omega - \mu_\omega[g])^2\, \tilde{\rho}_g(\omega)\, d\omega$$

[4]Ingrid Daubechies. "The Wavelet Transform, Time-Frequency Localization and Signal Analysis." *IEEE Transactions on Information Theory*, vol. 36, 1990, pp. 961–1005. https://doi.org/10.1109/18.57199.

3.6 The Time–Frequency Uncertainty Product

Finally, we combine time and frequency spread[5], with the following inequality applying regardless of the distribution of uncertainty between frequency and time, as demonstrated in Figure 2[6].

$$U[g] = \sigma_t[g] \cdot \sigma_\omega[g] \geq \frac{1}{2}$$

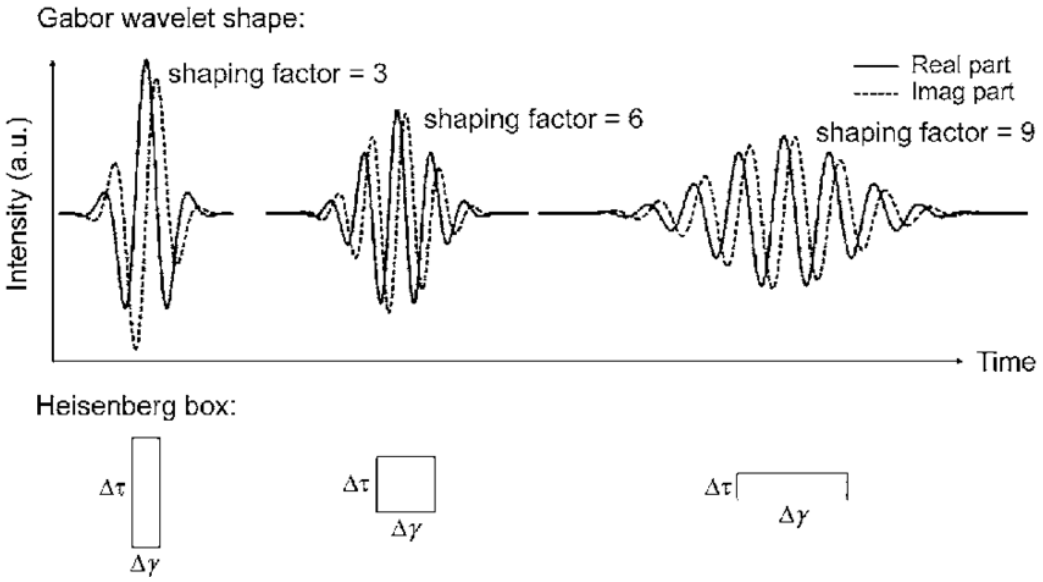


Figure 2: Heisenberg Uncertainty Principle in Time-Frequency Space

This inequality is the mathematical tool we will use to compare STFT and CWT.[7] Lower $U[g]$ means better simultaneous time-frequency concentration.

[5]———. "The Wavelet Transform, Time-Frequency Localization and Signal Analysis." *IEEE Transactions on Information Theory*, vol. 36, 1990, pp. 961–1005. https://doi.org/10.1109/18.57199.

[6]Yinbin Deng, et al. "Determination of Gabor Wavelet Shaping Factor for Accurate Phase Retrieval with Wavelet Transform." *Applied Physics B*, vol. 81, Dec. 2005, pp. 1107–11. https://doi.org/10.1007/s00340-005-1959-2.

[7]Matt Hall. "Resolution and Uncertainty in Spectral Decomposition." *First Break*,

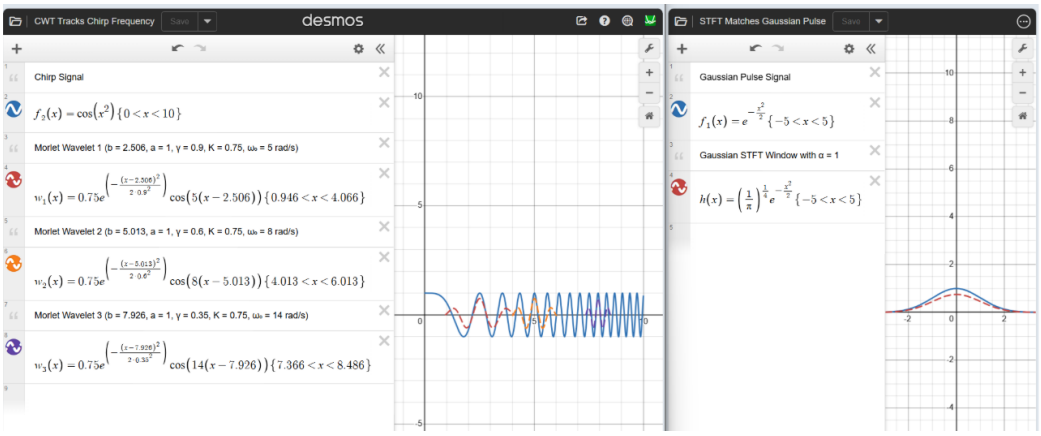


Figure 3: Expectation for Research Question

As shown in Figure 3[8], we expect STFT to perform well for the Gaussian pulse because its Gaussian window matches the signal's shape (both achieve $U = 1/2$). For the chirp, CWT may perform better as its adaptive scaling can track changing frequency better than STFT's fixed window.

In order to answer this question rigorously, this essay focuses on Gaussian-shaped basis functions for both transforms: a Gaussian window for STFT, and a Morlet wavelet (a Gaussian envelope modulated by a complex exponential) for CWT. Gaussians are the only functions that achieve the minimum time–frequency uncertainty product, so they represent the best-case resolution possible for each transform.

vol. 24, Dec. 2006. https://doi.org/10.3997/1365-2397.2006027.

[8]Desmos. "Desmos Graphing Calculator," www.desmos.com/calculator. Accessed 8 Oct. 2026.

4 STFT Uncertainty for Fixed Windows

4.1 Windowed Energy Conservation

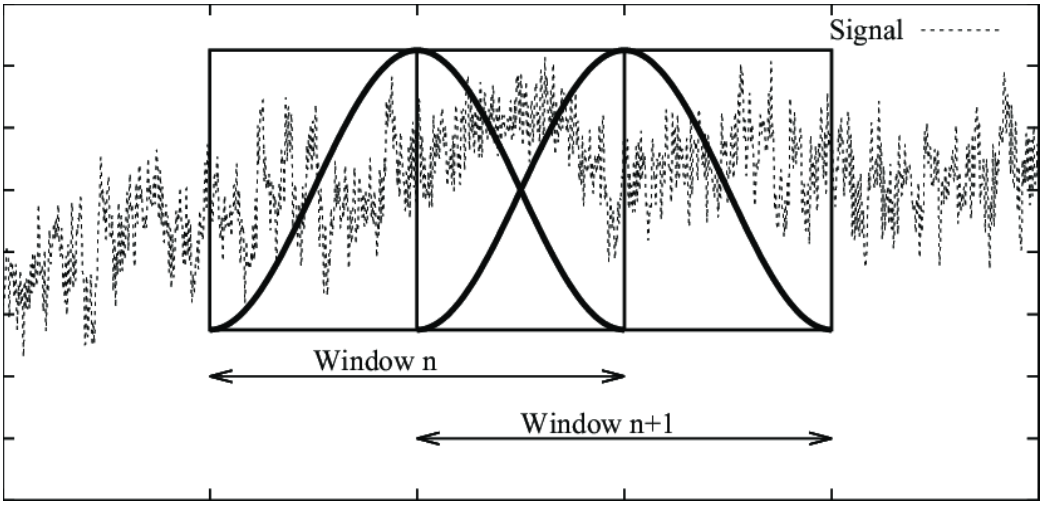


Figure 4: Short-Time Fourier Transform With Overlapping Gaussian Windows

As in Figure 4[9], the Short-Time Fourier Transform analyzes a signal $f(t)$ by multiplying it with a window function $h(t)$ localized in time. At each time shift τ, define the windowed signal $w_\tau(t) = f(t) \cdot h(t-\tau)$ and the STFT as:

$$STFT_f(\tau, \omega) = \int_{-\infty}^{\infty} f(t)h(t-\tau)e^{-i\omega t}dt$$

Since $w_\tau(t) = f(t)h(t-\tau)$, the STFT is simply the Fourier transform of $w_\tau(t)$.

[9]Konrad Kamieniecki, et al. "Basilar Membrane Vibration in Time Domain Predicted by Fluid–Structure Interaction Model in Pre- and Post-Stapedotomy State." *Procedia IUTAM*, vol. 24, 2017, pp. 48–63. https://doi.org/10.1016/j.piutam.2017.08.042.

Applying Parseval's theorem:

$$\int_{-\infty}^{\infty} |w_\tau(t)|^2 dt = \int_{-\infty}^{\infty} |f(t)|^2 |h(t-\tau)|^2 dt = \frac{1}{2\pi} \int_{-\infty}^{\infty} |STFT_f(\tau, \omega)|^2 d\omega$$

Thus energy is conserved between time and frequency domains for each time shift τ.

4.2 Variance Measures for the Window Function

Since STFT's resolution is determined by the window function, we compute its uncertainty product by applying the framework from Section 3. For window $h(t)$ with energy $E_h = \int_{-\infty}^{\infty} |h(t)|^2 dt$ and Fourier transform $H(\omega) = \int_{-\infty}^{\infty} h(t) e^{-i\omega t} dt$, the time and frequency variances are hence:

$$\sigma_t^2[h] = \frac{1}{E_h} \int_{-\infty}^{\infty} (t - \mu_t[h])^2 |h(t)|^2 dt, \quad \sigma_\omega^2[h] = \frac{1}{2\pi E_h} \int_{-\infty}^{\infty} (\omega - \mu_\omega[h])^2 |H(\omega)|^2 d\omega$$

where $\mu_t[h] = \frac{1}{E_h} \int_{-\infty}^{\infty} t|h(t)|^2 dt$ and $\mu_\omega[h] = \frac{1}{2\pi E_h} \int_{-\infty}^{\infty} \omega |H(\omega)|^2 d\omega$.

4.3 Relationship Between Window Parameters and Uncertainty

Since Gaussians are the only functions attaining the minimum uncertainty product, we will focus on Gaussian STFT Windows, describing the best possible time–frequency resolution that STFT can ever achieve.

We start with the general Gaussian shape $e^{-t^2/(2\alpha^2)}$, where $\alpha > 0$ is a width

parameter that controls how wide or narrow the window is.

4.3.1 Find the normalization constant

We need to make the window have unit energy ($E_h = 1$), such that extra constants will not be required when later working with the statistical mean and variance. Let the window be:

$$h(t) = C \cdot e^{-t^2/(2\alpha^2)}$$

where C is the normalization constant we must find.

Compute the energy:

$$E_h = \int_{-\infty}^{\infty} |h(t)|^2 dt = \int_{-\infty}^{\infty} C^2 e^{-t^2/\alpha^2} dt$$

Use the substitution $u = t/\alpha$, so $dt = \alpha du$:

$$E_h = C^2 \int_{-\infty}^{\infty} e^{-u^2} \cdot \alpha du = C^2 \alpha \int_{-\infty}^{\infty} e^{-u^2} du$$

The standard Gaussian integral is $\int_{-\infty}^{\infty} e^{-u^2} du = \sqrt{\pi}$. Therefore:

$$E_h = C^2 \alpha \sqrt{\pi}$$

Set $E_h = 1$ to enforce unit energy:

$$C^2 \alpha \sqrt{\pi} = 1 \implies C^2 = \frac{1}{\alpha\sqrt{\pi}} \implies C = \left(\frac{1}{\pi\alpha^2}\right)^{1/4}$$

Thus, the properly normalized Gaussian window is:

$$h(t) = \left(\frac{1}{\pi\alpha^2}\right)^{1/4} e^{-t^2/(2\alpha^2)}$$

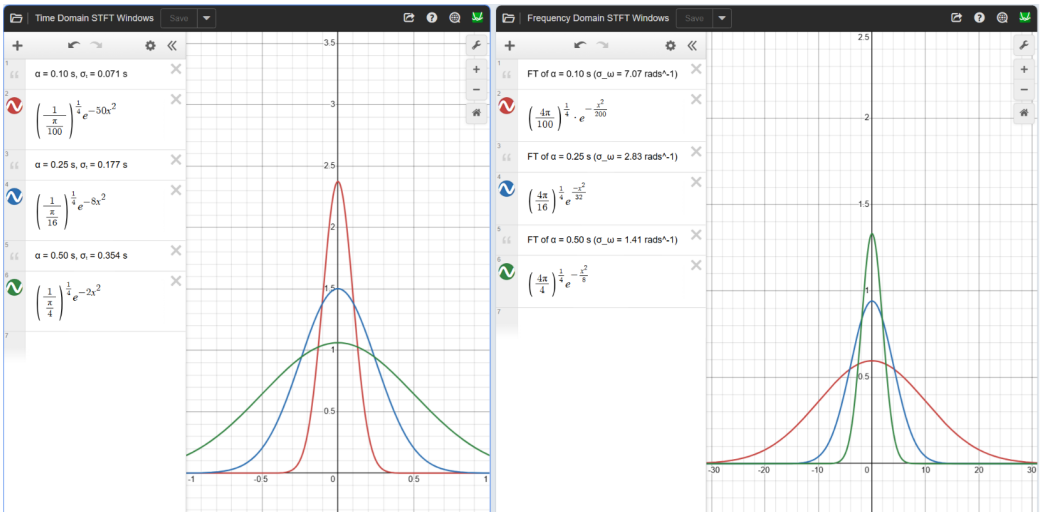


Figure 5: Varying of Gaussian Window Width with α

The parameter α controls the Gaussian window width[10] as shown in Figure 5[11], such that narrow windows have better time resolution and worse frequency resolution, and vice versa.[12]

[10]Shibli Nisar, et al. "An Efficient Adaptive Window Size Selection Method for Improving Spectrogram Visualization." *Computational Intelligence and Neuroscience*, vol. 2016, 2016, p. 6172453. https://doi.org/10.1155/2016/6172453.

[11]Desmos. "Desmos Graphing Calculator," www.desmos.com/calculator. Accessed 8 Oct. 2026.

[12]Carlos Mateo and Juan Antonio Talavera. "Short-Time Fourier Transform with the Window Size Fixed in the Frequency Domain." *Digital Signal Processing*, vol. 77, June 2018, pp. 13–21. https://doi.org/10.1016/j.dsp.2017.11.003.

4.3.2 Compute the Window Uncertainty Product

We can use the same probability density function framework, with $\rho_h(t) = |h(t)|^2$ ($h(t)$ already has unit energy).

Computing time variance:

First, we find the mean time:

$$\mu_t[h] = \int_{-\infty}^{\infty} t \cdot |h(t)|^2 dt = \int_{-\infty}^{\infty} t \cdot \frac{1}{\sqrt{\pi}\alpha} e^{-t^2/\alpha^2} dt = 0$$

The integral is zero because the integrand is an odd function, as by taking the constant:

$$A = \frac{e^{\frac{1}{\alpha^2}}}{\alpha\sqrt{\pi}}$$

It is evident, by the definition of odd functions $-f(-x) = f(x)$ that:

$$-(A(-t)e^{-(-t)^2}) = Ate^{-t^2}$$

Now compute the time variance:

$$\sigma_t^2[h] = \int_{-\infty}^{\infty} (t - \mu_t[h])^2 |h(t)|^2 dt = \int_{-\infty}^{\infty} t^2 |h(t)|^2 dt = \int_{-\infty}^{\infty} t^2 \cdot \frac{1}{\sqrt{\pi}\alpha} e^{-t^2/\alpha^2} dt$$

Let $u = t/\alpha$, so $dt = \alpha du$ and $t^2 = \alpha^2 u^2$:

$$\sigma_t^2[h] = \int_{-\infty}^{\infty} \alpha^2 u^2 \cdot \frac{1}{\sqrt{\pi}\alpha} e^{-u^2} \cdot \alpha du = \frac{\alpha^2}{\sqrt{\pi}} \int_{-\infty}^{\infty} u^2 e^{-u^2} du$$

Derivation of $\int_{-\infty}^{\infty} u^2 e^{-u^2}\,du$

We use integration by parts: $\int a\frac{db}{du}\,du = ab - \int b\frac{da}{du}\,du$.

Let:

$$\begin{cases} a = u & \implies \frac{da}{du} = 1 \\ \frac{db}{du} = ue^{-u^2} & \implies b = -\frac{1}{2}e^{-u^2} \end{cases}$$

We know the second with the substitution $\chi = -u^2$, $d\chi = -2u\,du$:

$$\int e^{-u^2}\,du = -\frac{1}{2}\int e^{\chi}\,d\chi \; = -\frac{1}{2}e^{\chi} = -\frac{1}{2}e^{-u^2}$$

Now apply integration by parts:

$$\int_{-\infty}^{\infty} u^2 e^{-u^2}\,du = \left[-\frac{u}{2}e^{-u^2}\right]_{-\infty}^{\infty} - \int_{-\infty}^{\infty}\left(-\frac{1}{2}e^{-u^2}\right)du$$

Considering the boundary term $\left[-\frac{u}{2}e^{-u^2}\right]_{-\infty}^{\infty}$:

$$\left[-\frac{u}{2}e^{-u^2}\right]_{-\infty}^{\infty} = \lim_{b\to\infty}\left(-\frac{b}{2}e^{-b^2}\right) - \lim_{a\to-\infty}\left(-\frac{a}{2}e^{-a^2}\right)$$

The following is sufficient to prove this equals 0.

$$\lim_{u\to\infty} ue^{-u^2} = \lim_{u\to-\infty} ue^{-u^2} = 0$$

Let us show $\lim_{u\to\pm\infty} u^2e^{-u^2} = 0$. Rewriting $u^2e^{-u^2}$ as $e^{\ln(u^2)-u^2}$, it becomes evident that as u approaches $\pm\infty$, u^2 grows much faster than $\ln(u^2)$, implying that $\ln(u^2) - u^2$ approaches $-\infty$. This signifies that $u^2e^{-u^2} = e^{\ln(u^2)-u^2}$ approaches 0.

For $|u| \geq 1$, we have:

$$0 \leq |u|e^{-u^2} \leq u^2 e^{-u^2}$$

From the argument above, we know $\lim_{u\to\pm\infty} u^2 e^{-u^2} = 0$, and clearly the lower bound is 0, hence by the squeeze theorem it is evident that.

$$\lim_{u\to\pm\infty} |u|e^{-u^2} = 0$$

Which means that the boundary term equals 0, leaving us with:

$$\int_{-\infty}^{\infty} u^2 e^{-u^2} du = \frac{1}{2}\int_{-\infty}^{\infty} e^{-u^2} du$$

The remaining integral is the standard Gaussian integral, which equals $\sqrt{\pi}$. Therefore:

$$\int_{-\infty}^{\infty} u^2 e^{-u^2} du = \frac{1}{2}\sqrt{\pi} = \frac{\sqrt{\pi}}{2}$$

Hence, as $\int_{-\infty}^{\infty} u^2 e^{-u^2} du = \frac{\sqrt{\pi}}{2}$:

$$\sigma_t^2[h] = \frac{\alpha^2}{\sqrt{\pi}}\int_{-\infty}^{\infty} u^2 e^{-u^2} du = \frac{\alpha^2}{\sqrt{\pi}} \cdot \frac{\sqrt{\pi}}{2} = \frac{\alpha^2}{2}$$

Thus, $\sigma_t[h] = \frac{\alpha}{\sqrt{2}}$.

Computing frequency variance:

To find the frequency variance of the Gaussian window, let us take the continuous Fourier transform of $h(t) = C \cdot e^{-t^2/(2\alpha^2)}$ where $C = (\pi\alpha^2)^{-1/4}$, with the

convention as:

$$H(\omega) = \int_{-\infty}^{\infty} h(t) e^{-i\omega t}\, dt$$

So:

$$H(\omega) = C \int_{-\infty}^{\infty} e^{\frac{-t^2}{2\alpha^2} - i\omega t}\, dt$$

Which can be written as:

$$H(\omega) = C \int_{-\infty}^{\infty} \exp\left(\frac{-t^2}{2\alpha^2} - i\omega t\right)\, dt$$

Taking the exponent, we have:

$$\frac{-t^2}{2\alpha^2} - i\omega t = -\frac{1}{2\alpha^2}(t^2 + 2i\alpha^2\omega t)$$

Taking $m = i\alpha^2\omega$:

$$= -\frac{1}{2\alpha^2}(t^2 + 2mt) = -\frac{1}{2\alpha^2}((t+m)^2 - m^2) = -\frac{1}{2\alpha^2}((t + i\alpha^2\omega)^2 - (-\alpha^4\omega^2))$$

$$= -\frac{1}{2\alpha^2}((t + i\alpha^2\omega)^2 + \alpha^4\omega^2) = \frac{-(t + i\alpha^2\omega)^2}{2\alpha^2} - \frac{\alpha^2\omega^2}{2}$$

As the $\frac{\alpha^2\omega^2}{2}$ term is constant with respect to t, we have:

$$H(\omega) = Ce^{-\alpha^2\omega^2/2} \int_{-\infty}^{\infty} \exp\left(-\frac{(t + i\alpha^2\omega)^2}{2\alpha^2}\right)\, dt$$

Let us substitute $\zeta = t + i\alpha^2\omega$ into the integral, where $d\zeta = dt$. Thus, with the standard Gaussian integral, we have:

$$\int_{-\infty}^{\infty} \exp\left(-\frac{\zeta^2}{2\alpha^2}\right) d\zeta = \sqrt{\frac{\pi}{\frac{1}{2\alpha^2}}} = \sqrt{2\pi\alpha^2} = \alpha\sqrt{2\pi}$$

Hence, we have the complete Continuous Fourier transform of the Gaussian window:

$$H(\omega) = C\alpha\sqrt{2\pi}e^{-\alpha^2\omega^2/2} = (4\alpha^4\pi^2)^{1/4}\left(\frac{1}{\pi\alpha^2}\right)^{1/4} e^{-\alpha^2\omega^2/2} = (4\pi\alpha^2)^{1/4}e^{-\alpha^2\omega^2/2}$$

As such, the normalized energy density in the frequency domain, as per our framework, is:

$$\tilde{\rho}_h(\omega) = \frac{|H(\omega)|^2}{2\pi E_h} = \frac{(4\pi\alpha^2)^{1/2}e^{-\alpha^2\omega^2}}{2\pi} = \frac{\alpha}{\sqrt{\pi}}e^{-\alpha^2\omega^2}$$

The mean frequency is $\mu_\omega[h] = 0$ by symmetry. The frequency variance is:

$$\sigma_\omega^2[h] = \int_{-\infty}^{\infty} \omega^2 \cdot \tilde{\rho}_h(\omega)d\omega = \int_{-\infty}^{\infty} \omega^2 \cdot \frac{\alpha}{\sqrt{\pi}}e^{-\alpha^2\omega^2}d\omega$$

Let $v = \alpha\omega$, so $\omega = v/\alpha$ and $d\omega = dv/\alpha$. We know $\int_{-\infty}^{\infty} x^2e^{-x^2}dx = \sqrt{\pi}/2$.

$$\sigma_\omega^2[h] = \int_{-\infty}^{\infty} \frac{v^2}{\alpha^2} \cdot \frac{\alpha}{\sqrt{\pi}}e^{-v^2} \cdot \frac{dv}{\alpha} = \frac{1}{\alpha^2\sqrt{\pi}}\int_{-\infty}^{\infty} v^2e^{-v^2}dv = \frac{1}{\alpha^2\sqrt{\pi}} \cdot \frac{\sqrt{\pi}}{2} = \frac{1}{2\alpha^2}$$

Thus, $\sigma_\omega[h] = \sqrt{\frac{1}{2\alpha^2}} = \frac{1}{\alpha\sqrt{2}}$.

4.4 The STFT Uncertainty Product

Combining our results:

$$U[h] = \sigma_t[h] \cdot \sigma_\omega[h] = \left(\frac{\alpha}{\sqrt{2}}\right) \cdot \left(\frac{1}{\alpha\sqrt{2}}\right) = \frac{1}{2}.$$

The Gaussian window therefore achieves the theoretical minimum uncertainty product $U[h] = 1/2$, regardless of the width parameter α. This result does *not* hold for arbitrary windows: commonly used alternatives such as rectangular, Hann, or Hamming windows all satisfy $U[h] > 1/2$. Thus, the Gaussian window describes the best-case time–frequency resolution achievable by STFT. In later sections, this value $U[h] = 1/2$ will be used as the resolution benchmark when STFT is applied to the Gaussian pulse and chirp.

Changing α stretches the window in time and squeezes it in frequency, or vice versa, but always in such a way that the product $\sigma_t[h]\sigma_\omega[h]$ stays constant.

The Gaussian window therefore achieves the theoretical minimum uncertainty product $U[h] = 1/2$, regardless of the width parameter α. Other common windows (Hamming, Hann, rectangular) yield $U[h] > 1/2$. Changing α adjusts the time-frequency trade-off while maintaining constant product.

5 CWT Analysis: Uncertainty of Scaled Wavelets

Now, we apply the finite-energy and variance framework to scaled wavelets $\psi_{a,b}(t)$ in CWT, and compute the uncertainty product for a specific mother wavelet.

5.1 Morlet Wavelet and CWT Definition

The Morlet wavelet[13] is defined by

$$\psi(t) = K\, e^{-t^2/(2\gamma^2)} e^{i\omega_0 t},$$

where $\gamma > 0$ controls the width of the Gaussian envelope in time, ω_0 is the central angular frequency, and K is a normalization constant chosen so that ψ has unit energy.

The Continuous Wavelet Transform of a signal $f(t)$ with respect to this mother wavelet is

$$W_f(a,b) = \frac{1}{\sqrt{|a|}} \int_{-\infty}^{\infty} f(t)\, \psi^*\left(\frac{t-b}{a}\right)\, dt,$$

where a is the scale and b the time shift, with wavelet scaling and shifting visually shown in Figure 6[14]. The corresponding analysing wavelet at scale a and position b is

$$\psi_{a,b}(t) = \frac{1}{\sqrt{|a|}} \psi\left(\frac{t-b}{a}\right).$$

The scalogram $S_f(a,b) = |W_f(a,b)|^2$ will later be interpreted using the uncertainty properties derived below.

[13] Ingrid Daubechies. "The Wavelet Transform, Time-Frequency Localization and Signal Analysis." *IEEE Transactions on Information Theory*, vol. 36, 1990, pp. 961–1005. https://doi.org/10.1109/18.57199.

[14] Petros Bogiatzis. "Using Wavelet Decomposition to Retrieve Information from Seismograms," 2026, www.seismology.harvard.edu/research/wavePicking.html. Accessed 10 Feb. 2026.

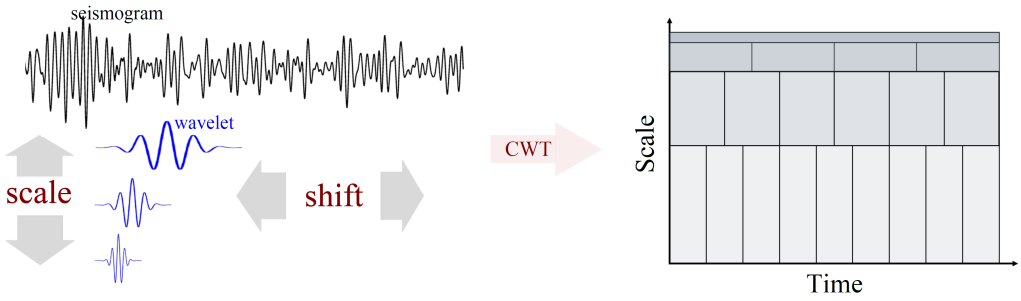


Figure 6: Wavelet Scaling and Shifting in CWT

5.2 Time-Domain Energy and Variance of the Morlet Wavelet

We first normalise ψ to unit energy:

$$E_\psi = \int_{-\infty}^{\infty} |\psi(t)|^2 dt = \int_{-\infty}^{\infty} |K|^2 e^{-t^2/\gamma^2} |e^{i\omega_0 t}|^2 dt = |K|^2 \int_{-\infty}^{\infty} e^{-t^2/\gamma^2} dt.$$

Since $|e^{i\omega_0 t}|^2 = 1$, the modulation does not change the energy envelope. With $u = t/\gamma$, $dt = \gamma du$,

$$E_\psi = |K|^2 \gamma \int_{-\infty}^{\infty} e^{-u^2} du = |K|^2 \gamma \sqrt{\pi}.$$

Setting $E_\psi = 1$ gives $|K|^2 = 1/(\gamma\sqrt{\pi})$.

The time-domain energy density is

$$\rho_\psi(t) = |\psi(t)|^2 = |K|^2 e^{-t^2/\gamma^2}.$$

By symmetry, $\rho_\psi(t)$ is an even function, so

$$\mu_t[\psi] = \int_{-\infty}^{\infty} t\,\rho_\psi(t)dt = 0.$$

The time variance is

$$\sigma_t^2[\psi] = \int_{-\infty}^{\infty} t^2 \rho_\psi(t)dt = |K|^2 \int_{-\infty}^{\infty} t^2 e^{-t^2/\gamma^2} dt.$$

With $u = t/\gamma$, $t = \gamma u$, $dt = \gamma du$, $t^2 = \gamma^2 u^2$,

$$\sigma_t^2[\psi] = |K|^2 \gamma^3 \int_{-\infty}^{\infty} u^2 e^{-u^2} du.$$

Using the integration-by-parts derivation from Section 4,

$$\int_{-\infty}^{\infty} u^2 e^{-u^2} du = \frac{\sqrt{\pi}}{2},$$

so

$$\sigma_t^2[\psi] = |K|^2 \gamma^3 \cdot \frac{\sqrt{\pi}}{2} = \frac{1}{\gamma\sqrt{\pi}} \gamma^3 \cdot \frac{\sqrt{\pi}}{2} = \frac{\gamma^2}{2},$$

and therefore

$$\sigma_t[\psi] = \frac{\gamma}{\sqrt{2}}.$$

5.3 Frequency-Domain Variance of the Morlet Wavelet

The Fourier transform of $\psi(t)$ can be computed by completing the square in the exponent, similar to the Gaussian window derivation. The result is a Gaussian in

frequency centred at ω_0:

$$\Psi(\omega) = K' e^{-(\gamma^2/2)(\omega-\omega_0)^2},$$

for some complex constant K' whose magnitude is fixed by the same energy normalization.

Thus

$$|\Psi(\omega)|^2 = |K'|^2 e^{-\gamma^2(\omega-\omega_0)^2}.$$

The frequency-domain energy density is

$$\tilde{\rho}_\psi(\omega) = \frac{|\Psi(\omega)|^2}{2\pi}.$$

By symmetry around ω_0,

$$\mu_\omega[\psi] = \omega_0.$$

Introducing $v = \gamma(\omega - \omega_0)$, so $\omega - \omega_0 = v/\gamma$, $d\omega = dv/\gamma$,

$$\sigma_\omega^2[\psi] = \int_{-\infty}^{\infty} (\omega-\omega_0)^2 \tilde{\rho}_\psi(\omega) d\omega = \int_{-\infty}^{\infty} \left(\frac{v}{\gamma}\right)^2 \frac{|K'|^2}{2\pi} e^{-v^2} \frac{dv}{\gamma}.$$

This is proportional to

$$\frac{1}{\gamma^2} \int_{-\infty}^{\infty} v^2 e^{-v^2} dv,$$

which again evaluates to $\sqrt{\pi}/2$ up to constants. Carrying through the normalization (in the same way as for the time variance) gives

$$\sigma_\omega^2[\psi] = \frac{1}{2\gamma^2}, \quad \text{so} \quad \sigma_\omega[\psi] = \frac{1}{\sqrt{2}\,\gamma}.$$

5.4 Uncertainty Product and Scaling Behaviour

For the mother wavelet itself we obtain

$$U[\psi] = \sigma_t[\psi]\sigma_\omega[\psi] = \left(\frac{\gamma}{\sqrt{2}}\right)\left(\frac{1}{\sqrt{2}\,\gamma}\right) = \frac{1}{2}.$$

Thus, like the Gaussian STFT window, the Morlet wavelet attains the minimum possible time–frequency uncertainty product allowed by the uncertainty principle.

Finally, consider the scaled wavelets

$$\psi_{a,b}(t) = \frac{1}{\sqrt{|a|}}\psi\left(\frac{t-b}{a}\right).$$

Repeating the same change-of-variables arguments with $u = (t-b)/a$ shows that their time and frequency standard deviations satisfy

$$\sigma_t[\psi_{a,b}] = |a|\,\sigma_t[\psi], \qquad \sigma_\omega[\psi_{a,b}] = \frac{1}{|a|}\,\sigma_\omega[\psi],$$

so for every scale a and shift b,

$$U[\psi_{a,b}] = \sigma_t[\psi_{a,b}]\sigma_\omega[\psi_{a,b}] = \sigma_t[\psi]\sigma_\omega[\psi] = \frac{1}{2}.$$

5.4.1 Interpretation of this Uncertainty Product

Both STFT and CWT achieve $U = \frac{1}{2}$. The crucial difference: STFT maintains this optimal trade-off fixed across all times and frequencies, whereas CWT redistributes

it scale-adaptively.[15] The Morlet wavelet is chosen for its Gaussian optimality; other wavelets (Mexican hat, Haar) yield $U > \frac{1}{2}$.

6 A Gaussian Pulse and Linear Chirp

We can now fairly test how the fixed optimal trade-off in STFT compares to the scale-adaptive optimal trade-off in CWT, knowing that ideally both can attain the minimum uncertainty product $U = \frac{1}{2}$ when analyzing two signals: Gaussian pulse and Linear chirp.

To make the comparison consistent, the spectrogram is the squared magnitude of the STFT, $|STFT_f(\tau, \omega)|^2$, displaying how energy is distributed over time shift τ and angular frequency ω. Similarly, the scalogram is the squared magnitude of the CWT, $|W_f(a, b)|^2$, showing energy distribution over time shift b and scale a.

6.1 Test Signals

Signal 1: Gaussian pulse

The Gaussian pulse is defined as $f_1(t) = e^{-t^2/2}$. This signal has one localized event in time centered at $t = 0$ and has no oscillation; its structure is primarily temporal localization.

[15] MathWorks. "Practical Introduction to Time-Frequency Analysis Using the Continuous Wavelet Transform," 2020, www.mathworks.com/help/wavelet/ug/practical-introduction-to-time-frequency-analysis-using-the-continuous-wavelet-transform.html. Accessed 1 Feb. 2026.

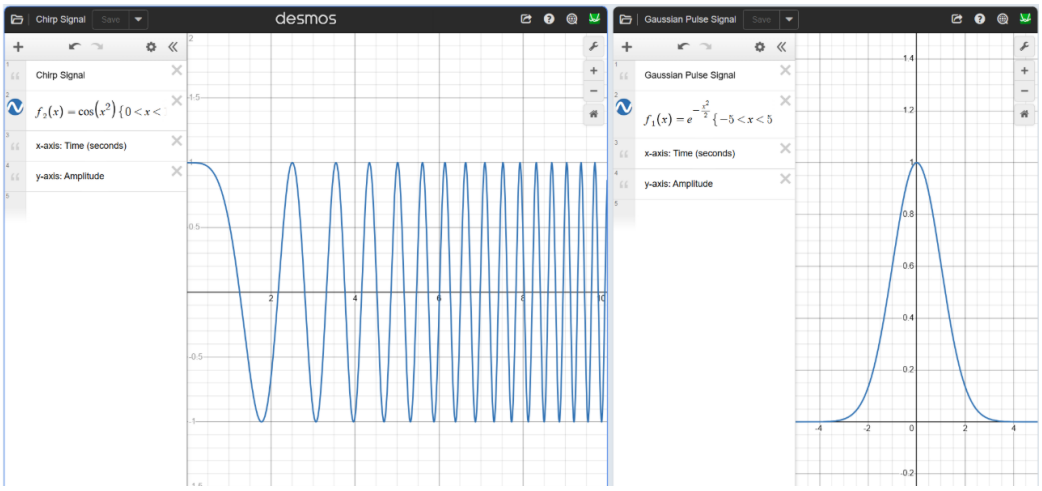


Figure 7: Gaussian Pulse and Linear Chirp[16]

Signal 2: Linear chirp

The chirp is defined as $f_2(t) = \cos(t^2)$, a cosine with time-varying phase $\Phi(t) = t^2$. The instantaneous angular frequency of a signal $\cos(\Phi(t))$ is found by differentiating the phase: $\omega_i(t) = \frac{d\Phi}{dt}$. Here $\Phi(t) = t^2$, so $\omega_i(t) = 2t$. Thus the chirp's instantaneous angular frequency increases linearly with time. This non-stationary behavior, where frequency changes continuously, is precisely the case where a fixed window can smear energy, motivating wavelet analysis.[17]

6.2 STFT on Both Signals

Recall the STFT definition from Section 4: $STFT_f(\tau, \omega) = \int_{-\infty}^{\infty} f(t)\, h(t-\tau)\, e^{-i\omega t}\, dt$, and the spectrogram is $\text{Spec}_f(\tau, \omega) = |STFT_f(\tau, \omega)|^2$.

[17]Pu Sun. "Comparison of STFT and Wavelet Transform in Time-Frequency Analysis," Dec. 2014, www.diva-portal.org/smash/get/diva2:793176/FULLTEXT01.pdf. Accessed 1 Feb. 2026.

Gaussian pulse under STFT

This investigation uses the Gaussian window $h(t) = \left(\frac{1}{\pi\alpha^2}\right)^{1/4} e^{-t^2/(2\alpha^2)}$, which achieves $U[h] = \frac{1}{2}$ for all α. For $f_1(t) = e^{-t^2/2}$, the signal is itself Gaussian-shaped, so the window is strongly matched to the signal: for τ near 0, the product $f_1(t)h(t-\tau)$ remains a smooth, single-peaked bump, and for large $|\tau|$ it rapidly decays. Therefore the spectrogram would show strong concentration near $\tau = 0$. Most spreading in ω is not due to oscillatory frequency content in the signal but due to the finite width of the windowed segment. As α decreases, the spectrogram becomes sharper in time and broader in frequency; as α increases, it becomes broader in time and sharper in frequency, while preserving the constant product $U[h] = \frac{1}{2}$. So, f_1 exemplifies a signal where STFT is optimal.

Linear chirp under STFT

For $f_2(t) = \cos(t^2)$, the windowed signal is $w_\tau(t) = \cos(t^2)\, h(t-\tau)$. Over a small time interval around $t = \tau$, the chirp behaves approximately like a sinusoid at frequency $\omega_i(\tau) = 2\tau$, but only if the window is narrow enough that $\omega_i(t)$ does not change significantly across the window. This is the key limitation of STFT: one fixed α must be used everywhere.

If α is chosen small, the STFT can track rapid changes in frequency because the window covers a shorter time range. However, $\sigma_\omega[h]$ becomes large, so the spectrogram's ridge in frequency becomes thick, resulting in poor frequency resolution. Conversely, if α is chosen large, frequency resolution improves, but the chirp changes frequency significantly across the window, so energy is smeared in time–frequency: the spectrogram does not follow a thin curve $\omega \approx 2\tau$ but spreads

around it. Thus, even though the Gaussian window is optimal in the uncertainty sense, the STFT representation of a chirp can still be visually and quantitatively smeared because the signal is not locally stationary at a single fixed time scale.

6.3 CWT on Both Signals

Recall the CWT definition from Section 5: $W_f(a,b) = \frac{1}{\sqrt{|a|}} \int_{-\infty}^{\infty} f(t)\, \psi^*\big(\frac{t-b}{a}\big)\, dt$, with analyzing wavelets $\psi_{a,b}(t) = \frac{1}{\sqrt{|a|}}\psi\big(\frac{t-b}{a}\big)$, and the time–scale scalogram $\mathrm{Scal}_f(a,b) = |W_f(a,b)|^2$. The chosen mother wavelet is the Morlet wavelet $\psi(t) = Ke^{-t^2/(2\gamma^2)}e^{i\omega_0 t}$. Sections 4–5 established that $U[\psi_{a,b}] = \frac{1}{2}$ for all a, b, with the scale relations $\sigma_t[\psi_{a,b}] = |a|\sigma_t[\psi]$ and $\sigma_\omega[\psi_{a,b}] = \frac{1}{|a|}\sigma_\omega[\psi]$.

Gaussian pulse under CWT

Since $f_1(t)$ is a single localized event, it contains broad frequency content but no sustained oscillation. The Morlet wavelet, however, is an oscillatory wave packet: a Gaussian envelope modulating a complex exponential. This means the CWT is not perfectly matched to a non-oscillatory pulse in the same direct way as a Gaussian STFT window is. Nevertheless, the CWT can still localize the pulse effectively in time because small scales a correspond to wavelets that are narrow in time, making them sensitive to sharp temporal localization, while large scales correspond to wavelets wider in time, which respond more weakly to a short pulse since the pulse occupies only a small fraction of the wavelet support. Therefore the scalogram would show strong energy concentration near $b = 0$ across scales, especially at scales whose wavelet time-width is comparable to the pulse width.

Linear chirp under CWT

For $f_2(t) = \cos(t^2)$, CWT's multi-resolution behavior means different scales naturally correspond to different frequency ranges, with small scales sensitive to higher frequencies and vice versa. Rather than STFT's fixed window width, CWT provides a family of analyzing functions $\psi_{a,b}$, where time and frequency spreads adjust with a. Thus, for a chirp whose instantaneous frequency increases with time, the scalogram would show energy concentrated along a curve: at early times with lower instantaneous frequency the dominant energy occurs at larger scales, and as time increases with higher instantaneous frequency the dominant energy shifts to smaller scales. As such, CWT's behavior is ideal for the chirp's varying frequencies.

6.4 Planned Numerical Procedure and Outputs

Answering the research question quantitatively requires the following stages.

Stage 1: Verify Basis Function Uncertainty

Sections 4 and 5 derived analytically that the Gaussian STFT window $h(t) = (\pi\alpha^2)^{-1/4}e^{-t^2/(2\alpha^2)}$ achieves $U[h] = \frac{1}{2}$ for all $\alpha > 0$, and that the Morlet wavelet $\psi(t) = Ke^{-t^2/(2\gamma^2)}e^{i\omega_0 t}$ achieves $U[\psi] = \frac{1}{2}$, with scaled versions $\psi_{a,b}(t)$ preserving $U[\psi_{a,b}] = \frac{1}{2}$ for all scales a and shifts b. For the validation of the analytical results, we must evaluate the variance integrals of Section 4 discretely, numerically computing $U[h]$ for the Gaussian Window for all $\alpha \in \{0.10, 0.25, 0.50\}$. This will similarly be done for $U[\psi_a]$ for the Morlet wavelet at scales $a \in \{0.5, 1.0, 2.0\}$ based on Section 5's variance formulas. As such, we can verify that all computed values satisfy $U \approx 0.5$, confirming that both transforms are using optimal basis functions.

This establishes, for our investigation, that STFT and CWT both have theoretical minimum uncertainty, meaning differences in performance are dependent on how this optimal resolution is distributed: fixed or scale-adaptive.

Stage 2: Signal-Specific Uncertainty Analysis

Next, we will analyse the time-frequency uncertainty when both STFT and CWT are applied to our two signals. First, to understand the signals' inherent properties, we will find the signal intrinsic uncertainties $U[f_1]$ and $U[f_2]$. Second, for each combination of signal, transform, and parameter, our variance framework will be applied to obtain U_{rep}, essentially representing how localized the time and frequency are in each spectogram and scalogram. The STFT window width parameter is $\alpha \in \{0.10, 0.25, 0.50\}$ s, and for CWT the Morlet wavelet across 3 scales from 2–20 Hz.

For the Gaussian pulse, STFT would perform well due to shape-matching with the Gaussian window. For the chirp, it is unclear whether CWT's scale-adaptive resolution or STFT's shape-matched optimal window will dominate. CWT may track frequency variation better, but STFT's Gaussian optimality might compensate. The numerical investigation will resolve this ambiguity. Comparing these numerical values will provide a direct, quantitative answer to the research question.

7 Computational Implementation

This section presents the numerical calculations that validate the analytical results from Sections 4–5 and answer the research question quantitatively.

7.1 Discretization Method

The continuous integrals derived analytically, including time and frequency variance, must be approximated using discrete samples for computer calculation. The standard **Riemann sum approximation** replaces an integral with a finite sum:

$$\int_a^b f(t)dt \approx \sum_{n=1}^{N} f(t_n)\Delta t,$$

where t_n are equally spaced points with spacing $\Delta t = (b-a)/N$.

The following sampling parameters were used: time resolution $\Delta t = 0.0025$ s (sampling frequency $f_s = 400$ Hz), Gaussian pulse on $t \in [-5, 5]$ s with $N = 4000$ samples, chirp on $t \in [0, 10]$ s with $N = 4000$ samples, and basis functions on $t \in [-10, 10]$ s with $N = 8000$ samples. The Gaussian pulse interval ensures $e^{-t^2/2} < 10^{-10}$ at boundaries, making the truncation error negligible.

Discrete time variance

For discrete samples $g[n]$ at times t_n, the time-domain energy is $E_g \approx \sum_{n=1}^{N} |g[n]|^2 \Delta t$, the normalized probability density is $\rho_g[n] = |g[n]|^2/E_g$, the mean is $\mu_t[g] \approx \sum_{n=1}^{N} t_n \cdot \rho_g[n]\Delta t$, and the variance is $\sigma_t^2[g] \approx \sum_{n=1}^{N}(t_n - \mu_t)^2 \cdot \rho_g[n]\Delta t$.

Discrete frequency variance

The Discrete Fourier Transform (DFT)[18] computes the frequency representation: $G[k] = \sum_{n=0}^{N-1} g[n]e^{-2\pi ikn/N}$. Each index k corresponds to angular frequency $\omega_k =$

[18]Project Nayuki. "How to Implement the Discrete Fourier Transform," June 2018, www.nayuki.io/page/how-to-implement-the-discrete-fourier-transform. Accessed 1 Feb. 2026.

$2\pi k/(N\Delta t)$ rad/s, with frequency spacing $\Delta\omega = 2\pi/(N\Delta t)$. The normalized frequency-domain energy density is $\tilde{\rho}_g[k] = |G[k]|^2 \cdot \Delta\omega/(2\pi E_g)$, and the frequency variance is computed as $\sigma_\omega^2[g] \approx \sum_k (\omega_k - \mu_\omega)^2 \cdot \tilde{\rho}_g[k]\Delta\omega$.

Two-dimensional variance for spectrograms and scalograms

Both the STFT spectrogram $P_{\text{STFT}}[m,n] = |\text{STFT}[\tau_m, \omega_n]|^2$ and CWT scalogram $P_{\text{CWT}}[m,n] = |\text{CWT}[a_m, b_n]|^2$ (converted to frequency via scale-to-frequency mapping) are normalized as $P_{\text{norm}}[m,n] = P[m,n]/(\sum_{m,n} P[m,n]\Delta\tau\Delta f)$. The time and frequency variances are then computed identically for both transforms: $\sigma_t^2 = \sum_{m,n}(\tau_m - \mu_t)^2 P_{\text{norm}}[m,n]\Delta\tau\Delta f$ and $\sigma_f^2 = \sum_{m,n}(f_n - \mu_f)^2 P_{\text{norm}}[m,n]\Delta\tau\Delta f$, where $\sigma_\omega = 2\pi\sigma_f$ converts to angular frequency. These compute the spread of energy across the time-frequency plane, treating the representation as a two-dimensional probability distribution.[19]

7.2 Parameter Selection Rationale

Both transforms are tested with 3 distinct parameters, varying resolution.

STFT window width selection

The window width α controls time-frequency trade-off: smaller α improves time resolution but worsens frequency resolution. We test $\alpha \in \{0.10, 0.25, 0.50\}$ s. The narrow window ($\alpha = 0.10$ s, $\sigma_t \approx 0.071$ s) emphasizes time localization, covering 5% of the pulse duration. The medium window ($\alpha = 0.25$ s, $\sigma_t \approx 0.177$ s) balances

[19]The Moonlight. "Recall Gabor Communication Theory and Joint Time-Frequency Analysis," Oct. 2025, Literature review, www.themoonlight.io/en/review/recall-gabor-communication-theory-and-joint-time-frequency-analysis. Accessed 1 Feb. 2026.

both domains. The wide window ($\alpha = 0.50$ s, $\sigma_t \approx 0.354$ s) emphasizes frequency resolution.[20]

CWT frequency band selection

Similar to STFT's scaling with window width α, by varying the Morlet wavelet's scale a, representing different frequency bands, we can test CWT analogously.

Firstly, unlike STFT, CWT should naturally be scaled with frequencies that are logarithmically spaced instead due to the geometric nature of wavelets.

Secondly, the Gaussian pulse has the most energy in 0-2 Hz, while the chirp ranges from 0-5 Hz.

Hence, we can test three frequency bands with ±30% range (15 scales each): a low band at 2 Hz (1.4–2.6 Hz) to match signal content; a middle band at 8 Hz (5.6–10.4 Hz) for intermediate resolution; a high band at 20 Hz (14–26 Hz) to emphasize time localization.[21]

7.3 Stage 1: Basis Function Verification

Upon implementing a Python script for calculating the uncertainties of both basis functions[22], it is clear both achieve $U \approx 0.5$, as per the analytical results. The

[20] Shibli Nisar, et al. "An Efficient Adaptive Window Size Selection Method for Improving Spectrogram Visualization." *Computational Intelligence and Neuroscience*, vol. 2016, 2016, p. 6172453. https://doi.org/10.1155/2016/6172453.

[21] MathWorks. "Practical Introduction to Time-Frequency Analysis Using the Continuous Wavelet Transform," 2020, www.mathworks.com/help/wavelet/ug/practical-introduction-to-time-frequency-analysis-using-the-continuous-wavelet-transform.html. Accessed 1 Feb. 2026.

[22] Project Nayuki. "How to Implement the Discrete Fourier Transform," June 2018, www.nayuki.io/page/how-to-implement-the-discrete-fourier-transform. Accessed 1 Feb. 2026.

result of this discretization is shown in Tables 1 and 2.

α (s)	σ_t (s)	σ_ω (rad/s)	$U[h]$
0.10	0.0707	7.071	0.500
0.25	0.1768	2.828	0.500
0.50	0.3536	1.414	0.500

Table 1: Gaussian window uncertainty verification.

Scale a	σ_t (s)	σ_ω (rad/s)	$U[\psi_a]$
0.5	0.612	0.834	0.511
1.0	1.225	0.418	0.512
2.0	2.449	0.209	0.512

Table 2: Morlet wavelet uncertainty verification.

The Gaussian window achieves $U[h] = 0.500$ across all widths, with a computational error under 0.1%, relative to the analytical value. The Morlet wavelet achieves $U[\psi_a]$ between 0.511 and 0.512 across all three scales, meaning there is an error less than 2.5% relative to the theoretical value of 0.5. This small discrepancy arises from the use of finite sampling in Discrete Wavelet Transform approximations.

7.4 Stage 2A: Signal Intrinsic Uncertainty

Table 3 shows the intrinsic uncertainty for the test signals, calculated with the same Python framework from Stage 1. The Gaussian pulse achieves the optimal $U = 0.5$, while the chirp has $U \approx 34.3 \gg 0.5$ due to its time-varying frequency.

Signal	σ_t (s)	σ_ω (rad/s)	$U[f]$
Gaussian pulse	0.707	0.707	0.500
Chirp	2.991	11.467	34.30

Table 3: Intrinsic signal uncertainty.

7.5 Stage 2B: Transform Concentration

By developing a Python script for applying both STFT and CWT to the Gaussian pulse, the parameters where both perform best can be found, as shown by the minimum U_{rep}, bolded in Tables 4 and 5. This fundamentally quantifies which transform achieves better concentration for each signal type.

Transform Configuration	σ_t (s)	σ_ω (rad/s)	U_{rep}
STFT ($\alpha = 0.10$ s)	0.711	4.770	3.389
STFT ($\alpha = 0.25$ s)	0.729	1.954	1.424
STFT ($\alpha = 0.50$ s)	**0.788**	**1.054**	**0.831**
CWT (20 Hz band)	1.225	23.298	28.540
CWT (8 Hz band)	1.226	4.574	5.607
CWT (2 Hz band)	**0.922**	**1.520**	**1.402**

Table 4: Transform concentration for Gaussian pulse.

STFT with $\alpha = 0.50$ s is evidently ideal for the Gaussian Pulse, with $U_{\text{rep}} = 0.831$ outperforming CWT's optimal $U_{\text{rep}} = 1.402$ by 69%. As previously hypothesized, this shows STFT's superiority for signals that are stationary. Figure 8 visualizes this difference through a spectogram and scalogram, built with MatPlotLib in Python. Since the Gaussian window and Gaussian pulse have matching shapes, STFT is tight concentrated. In contrast, CWT is broadly spread, as the oscillatory nature of the Morlet wavelet is not suitable for the stationary Gaussian pulse.

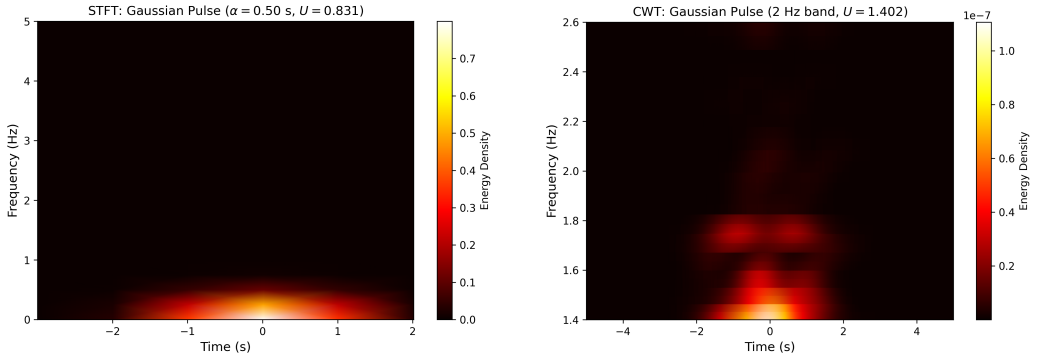


Figure 8: Optimal time-frequency representations of Gaussian pulse.

Transform Configuration	σ_t (s)	σ_ω (rad/s)	U_{rep}
STFT ($\alpha = 0.10$ s)	2.747	8.108	22.27
STFT ($\alpha = 0.25$ s)	2.294	5.558	12.75
STFT ($\alpha = 0.50$ s)	**1.723**	**3.850**	**6.632**
CWT (20 Hz band)	4.948	21.542	106.5
CWT (8 Hz band)	4.650	8.176	38.02
CWT (2 Hz band)	**1.499**	**2.315**	**3.471**

Table 5: Transform concentration for linear chirp.

CWT's optimal 2 Hz band achieves $U_{\text{rep}} = 3.471$, outperforming STFT's optimal $\alpha = 0.50$ s ($U_{\text{rep}} = 6.632$) by 91%. This demonstrates CWT's superiority for time-varying signals. Figure 9 shows STFT produces a thicker energy ridge because its fixed window cannot fully adapt to frequency variation, while CWT achieves tighter localization along the chirp trajectory through scale adaptivity.

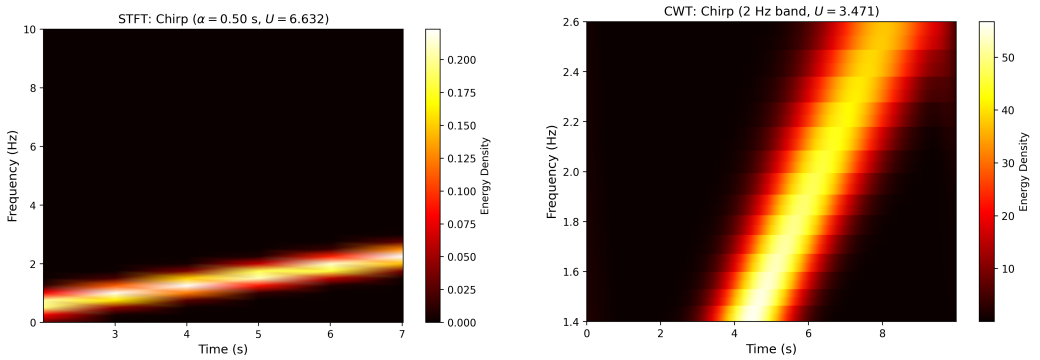


Figure 9: Optimal time-frequency representations of linear chirp.

7.6 Interpretation

General observations

All values of U_{rep} exceed the basis functions' minimum of 0.5 even with the windows or wavelets being optimal, even slight mismatches between signal characteristics and basis function properties can reduce energy concentration across both time and frequency. This representation uncertainty will approach the ideal 0.5 only if signal shape, frequency content, and basis function are all optimally well-aligned (unrealistic practically).

Figure 10 shows STFT's time-frequency trade-off for analyzing the chirp signal. The narrow window $\alpha = 0.10$ s led to vertical stripes in the spectogram, resulting from poor frequency resolution ($U = 22.27$). The medium window $\alpha = 0.25$ s provides better balance ($U = 12.75$). The wide window $\alpha = 0.50$ s gives optimal concentration with smoothest frequency tracking ($U = 6.63$), though still worse than CWT's scale-adaptive approach.

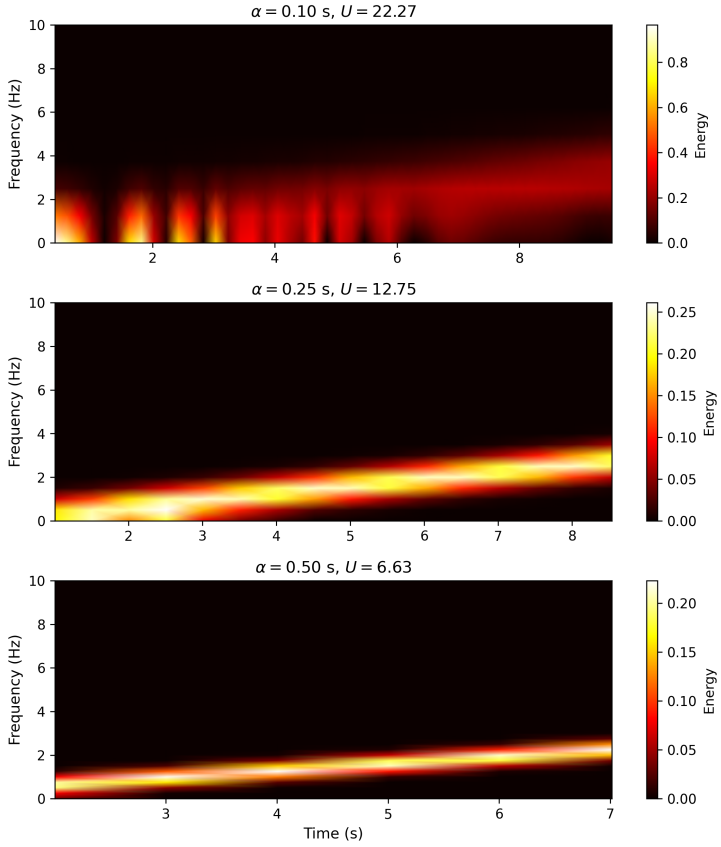


Figure 10: STFT parameter effects on chirp representation.

Gaussian pulse analysis

STFT with $\alpha = 0.50$ s achieved $U_{\text{rep}} = 0.831$, outperforming CWT's optimal 2 Hz band ($U_{\text{rep}} = 1.402$) by 1.69×. The wide Gaussian window matches both the signal's temporal extent and Gaussian shape. CWT performs worse because the oscillatory Morlet wavelet is mismatched to the non-oscillatory pulse structure.

Chirp analysis

CWT with 2 Hz band achieved $U_{\text{rep}} = 3.471$, outperforming STFT's optimal $\alpha = 0.50$ s ($U_{\text{rep}} = 6.63$) by 1.91×. The 2 Hz band matches the chirp's swept frequency range (0–3.2 Hz) and the wavelet's oscillatory structure aligns with the sinusoidal signal. CWT's scale adaptivity tracks time-varying frequency more effectively than STFT's fixed window.

8 Conclusion

This investigation addressed how the effectiveness of STFT versus CWT in achieving precise time-frequency resolution depend on whether the signal is stationary or time-varying in frequency. Through quantitative analysis of uncertainty products applied to a stationary Gaussian pulse and a non-stationary linear chirp signal, the investigation demonstrates that no single transform universally dominates; rather, effectiveness is determined by the match between signal characteristics and transform properties.

8.1 Answer to the Research Question

Both the Gaussian STFT window and Morlet wavelet achieve the theoretical minimum uncertainty product $U = 0.5$, confirming their optimality as basis functions.[23][24] However, their practical effectiveness diverges based on signal structure.

[23]Dennis Gabor. "Theory of Communication," Nov. 1944, jcsphysics.net/lit/gabor1946.pdf. Accessed 1 Feb. 2026.

[24]Ingrid Daubechies. "The Wavelet Transform, Time-Frequency Localization and Signal Analysis." *IEEE Transactions on Information Theory*, vol. 36, 1990, pp. 961–1005. https://doi.org/10.1109/18.57199.

For the stationary Gaussian pulse, STFT achieved $U_{\text{rep}} = 0.831$, outperforming CWT by 1.69×, demonstrating that shape-matching between the Gaussian window and Gaussian signal dominates performance. Conversely, for the non-stationary chirp, CWT achieved $U_{\text{rep}} = 3.471$, outperforming STFT by 1.91×, because CWT's scale-adaptive resolution better tracks the chirp's time-varying frequency and the oscillatory wavelet better matches the chirp's sinusoidal structure.[25][26]

Essentially, for stationary signals, STFT's Gaussian window shape-matching dominates. For time-varying signals, CWT's scale-adaptive frequency tracking dominates.

8.2 Practical Implications

These results have direct significance for signal processing applications where choosing the appropriate time-frequency analysis method is critical. For stationary, localized signals such as transient pulses in radar systems or isolated seismic events, STFT with appropriately sized Gaussian windows provides superior localization by matching the signal's temporal compactness. Conversely, for non-stationary signals with time-varying frequency content, such as chirp radar returns, frequency-modulated communications, or biological signals with evolving spectral characteristics, CWT's multi-scale analysis better tracks the instantaneous frequency variations. Essentially, the way the wavelet adapts its resolution

[25] Pu Sun. "Comparison of STFT and Wavelet Transform in Time-Frequency Analysis," Dec. 2014, www.diva-portal.org/smash/get/diva2:793176/FULLTEXT01.pdf. Accessed 1 Feb. 2026.

[26] Matt Hall. "Resolution and Uncertainty in Spectral Decomposition." *First Break*, vol. 24, Dec. 2006. https://doi.org/10.3997/1365-2397.2006027.

at different scales, providing better time resolution at high frequencies and superior frequency resolution at low frequencies, makes it innately suitable for analyzing signals whose frequency content changes over time.

8.3 Limitations and Future Directions

The first limitation was CWT frequency bands being chosen with logarithmic spacing starting at the signal's dominant frequency, meaning the derived optimal CWT configuration might only partly reflect this selection. Yet, for this investigations's scope, this selection was reasonable. Secondly, the achieved U_{rep} metric reduces a two-dimensional time-frequency spread to a single number, even though two representations with identical U_{rep} need not be identical. In the future, testing a broader sweep of different CWT frequency bands would allow us to see whether the performance advantage was biased by band selection. Further, comparing achieved U_{rep} against other, alternative concentration metrics would permit more informed conclusions, and reveal nuances in the applications of STFT and CWT. Also, more broadly, extending the analysis to other signals with noise or overlapping frequency components would test how the shape-matching and scale-adaptivity advantages seen here would survive in realistic settings.

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10 Appendix

10.1 Appendix A: Python Code for Calculating Uncertainty Products (Section 7)

```
import numpy as np
from scipy import signal, fft
import pywt


fs = 400.0
dt = 1.0 / fs


t_basis = np.linspace(-10, 10, 8000)
t_pulse = np.linspace(-5, 5, 4000)
f1 = np.exp(-(t_pulse**2)/2)


T_chirp = 10.0
t_chirp = np.arange(0.0, T_chirp, dt)
f2 = np.cos(t_chirp**2)


alphas = [0.10, 0.25, 0.50]
scales_test = [0.5, 1.0, 2.0]
wavelet = "cmor1.5-1.0"


cwt_scale_configs = [
```

```
    {"name": "narrow", "center_freq": 20, "bandwidth": 0.3},
    {"name": "medium", "center_freq": 8, "bandwidth": 0.3},
    {"name": "wide", "center_freq": 2, "bandwidth": 0.3}
]

def compute_1d_uncertainty(g, t):
    dt_val = np.mean(np.diff(t))

    rho_t = np.abs(g)**2
    E = np.sum(rho_t) * dt_val
    if E < 1e-12:
        return np.nan, np.nan, np.nan
    rho_t = rho_t / E

    mu_t = np.sum(t * rho_t) * dt_val
    sigma_t2 = np.sum((t - mu_t)**2 * rho_t) * dt_val
    sigma_t = np.sqrt(sigma_t2)

    G = fft.fftshift(fft.fft(fft.ifftshift(g)))
    N = len(g)
    omega = 2 * np.pi * fft.fftshift(fft.fftfreq(N, dt_val))

    rho_omega = np.abs(G)**2
    domega = np.mean(np.diff(omega))
```

```
    E_omega = np.sum(rho_omega) * domega / (2*np.pi)
    rho_omega = rho_omega / E_omega / (2*np.pi)

    mu_omega = np.sum(omega * rho_omega) * domega
    sigma_omega2 = np.sum((omega - mu_omega)**2 * rho_omega) *
    ↪  domega
    sigma_omega = np.sqrt(sigma_omega2)

    U = sigma_t * sigma_omega
    return sigma_t, sigma_omega, U

print("STAGE 1: Basis Function Uncertainty")
print("STFT Gaussian Window")
for alpha in alphas:
    C = (np.pi * alpha**2)**(-0.25)
    h = C * np.exp(-t_basis**2 / (2 * alpha**2))
    sigma_t, sigma_omega, U = compute_1d_uncertainty(h, t_basis)
    print(f" = {alpha:.2f} s: _t={sigma_t:.4f}s,
    ↪  _={sigma_omega:.4f}rad/s, U={U:.4f}")

print("\nMorlet Wavelet at different scales")
cwt_obj = pywt.ContinuousWavelet(wavelet)
psi_base, t_base = cwt_obj.wavefun(level=10)
t_base_scaled = t_base * 2.0
```

```
for a in scales_test:
    t_scaled = t_basis / a
    psi_real = np.interp(t_scaled, t_base_scaled,
    ↪  np.real(psi_base), left=0, right=0)
    psi_imag = np.interp(t_scaled, t_base_scaled,
    ↪  np.imag(psi_base), left=0, right=0)
    psi_a = (psi_real + 1j * psi_imag) / np.sqrt(np.abs(a))
    sigma_t, sigma_omega, U = compute_1d_uncertainty(psi_a,
    ↪  t_basis)
    print(f"Scale a = {a:.1f}: _t={sigma_t:.4f}s,
    ↪  _={sigma_omega:.4f}rad/s, U={U:.4f}")

print("STAGE 2A: Signal Intrinsic Uncertainty")
print("Gaussian pulse")
sigma_t, sigma_omega, U = compute_1d_uncertainty(f1, t_pulse)
print(f"  _t={sigma_t:.4f}s, _={sigma_omega:.4f}rad/s, U={U:.4f}")

print("\nChirp")
sigma_t, sigma_omega, U = compute_1d_uncertainty(f2, t_chirp)
print(f"  _t={sigma_t:.4f}s, _={sigma_omega:.4f}rad/s, U={U:.4f}")

def compute_2d_tf_concentration(P, t_axis, f_axis):
    dt_bin = np.mean(np.diff(t_axis)) if len(t_axis) > 1 else 1.0
```

```
    df_bin = np.mean(np.diff(f_axis)) if len(f_axis) > 1 else 1.0

    total_E = np.sum(P) * dt_bin * df_bin
    if total_E == 0:
        return np.nan, np.nan, np.nan
    P_norm = P / total_E

    mu_t = np.sum(P_norm * t_axis[None, :]) * dt_bin * df_bin
    mu_f = np.sum(P_norm * f_axis[:, None]) * dt_bin * df_bin

    sigma_t2 = np.sum(P_norm * (t_axis[None, :] - mu_t)**2) *
    ↪  dt_bin * df_bin
    sigma_f2 = np.sum(P_norm * (f_axis[:, None] - mu_f)**2) *
    ↪  dt_bin * df_bin

    sigma_t = np.sqrt(sigma_t2)
    sigma_omega = 2 * np.pi * np.sqrt(sigma_f2)
    return sigma_t, sigma_omega, sigma_t * sigma_omega

def gaussian_window(alpha_seconds, fs, n_std=4):
    half_len = int(np.ceil(n_std * alpha_seconds * fs))
    n = np.arange(-half_len, half_len + 1)
    tt = n / fs
    return np.exp(-(tt**2) / (2 * alpha_seconds**2))
```

```
print("STAGE 2B: Transform Representation Concentration")
print("Gaussian pulse:")
for alpha in alphas:
    win = gaussian_window(alpha, fs, n_std=4)
    f, t_seg, Zxx = signal.stft(f1, fs=fs, window=win,
    ↪  nperseg=len(win),
                                noverlap=int(0.75*len(win)),
                                ↪  boundary=None, padded=False)
    P = np.abs(Zxx)**2
    t_real = t_pulse[0] + t_seg
    sigma_t, sigma_omega, U = compute_2d_tf_concentration(P,
    ↪  t_real, f)
    print(f"  STFT (={alpha:.2f}s): _t={sigma_t:.4f}s,
    ↪  _={sigma_omega:.4f}rad/s, U={U:.4f}")

print("\n  CWT at different scale bands:")
for config in cwt_scale_configs:
    center_freq = config["center_freq"]
    bandwidth = config["bandwidth"]

    f_low = center_freq * (1 - bandwidth)
    f_high = center_freq * (1 + bandwidth)
```

```
    freqs_band = np.geomspace(f_low, f_high, 15)
    scales_band = pywt.frequency2scale(wavelet, freqs_band / fs)

    coefs, freqs_actual = pywt.cwt(f1, scales_band, wavelet,
    ↪  sampling_period=dt, method="fft")
    P_cwt = np.abs(coefs)**2

    sigma_t, sigma_omega, U = compute_2d_tf_concentration(P_cwt,
    ↪  t_pulse, freqs_actual)
    print(f"  CWT ({config['name']}, f{center_freq}Hz
    ↪  ±{int(bandwidth*100)}%, 15 scales):")
    print(f"       _t={sigma_t:.4f}s, _={sigma_omega:.4f}rad/s,
    ↪  U={U:.4f}")

print("Chirp:")
for alpha in alphas:
    win = gaussian_window(alpha, fs, n_std=4)
    f, t_seg, Zxx = signal.stft(f2, fs=fs, window=win,
    ↪  nperseg=len(win),
                                noverlap=int(0.75*len(win)),
                                ↪  boundary=None, padded=False)
    P = np.abs(Zxx)**2
    t_real = t_chirp[0] + t_seg
```

```
        sigma_t, sigma_omega, U = compute_2d_tf_concentration(P,
        ↪  t_real, f)
        print(f"  STFT (={alpha:.2f}s): _t={sigma_t:.4f}s,
        ↪  _={sigma_omega:.4f}rad/s, U={U:.4f}")

    print("\n  CWT at different scale bands:")
    for config in cwt_scale_configs:
        center_freq = config["center_freq"]
        bandwidth = config["bandwidth"]

        f_low = center_freq * (1 - bandwidth)
        f_high = center_freq * (1 + bandwidth)

        freqs_band = np.geomspace(f_low, f_high, 15)
        scales_band = pywt.frequency2scale(wavelet, freqs_band / fs)

        coefs, freqs_actual = pywt.cwt(f2, scales_band, wavelet,
        ↪  sampling_period=dt, method="fft")
        P_cwt = np.abs(coefs)**2

        sigma_t, sigma_omega, U = compute_2d_tf_concentration(P_cwt,
        ↪  t_chirp, freqs_actual)
        print(f"  CWT ({config['name']}, f{center_freq}Hz
        ↪  ±{int(bandwidth*100)}%, 15 scales):")
```

```
    print(f"      _t={sigma_t:.4f}s, _={sigma_omega:.4f}rad/s,
    ↪  U={U:.4f}")
```

10.2 Appendix B: Python Code for Spectograms and Scalograms (Section 7)

```
import numpy as np
import matplotlib.pyplot as plt
from scipy import signal
import pywt


fs = 400.0
dt = 1.0 / fs


t_pulse = np.linspace(-5, 5, 4000)
f1 = np.exp(-(t_pulse**2)/2)


t_chirp = np.arange(0.0, 10.0, dt)
f2 = np.cos(t_chirp**2)


print("Generating figures...")
alpha = 0.50
half_len = int(np.ceil(4 * alpha * fs))
n = np.arange(-half_len, half_len + 1)
tt = n / fs
```

```
win = np.exp(-(tt**2) / (2 * alpha**2))

f, t, Zxx = signal.stft(f1, fs=fs, window=win, nperseg=len(win),
                        noverlap=int(0.75*len(win)),
                        ↪  boundary=None, padded=False)

plt.figure(figsize=(7, 5))
plt.pcolormesh(t_pulse[0] + t, f, np.abs(Zxx)**2,
↪  shading='gouraud', cmap='hot')
plt.colorbar(label='Energy Density')
plt.xlabel('Time (s)', fontsize=12)
plt.ylabel('Frequency (Hz)', fontsize=12)
plt.title('STFT: Gaussian Pulse ($\\alpha=0.50$ s, $U=0.831$)',
↪  fontsize=12)
plt.ylim([0, 5])
plt.tight_layout()
plt.savefig('stft_gaussian_pulse.png', dpi=300,
↪  bbox_inches='tight')
print("Saved: stft_gaussian_pulse.png")
plt.close()

freqs_band = np.geomspace(1.4, 2.6, 15)
scales_band = pywt.frequency2scale('cmor1.5-1.0', freqs_band / fs)
```

```
coefs, freqs_actual = pywt.cwt(f1, scales_band, 'cmor1.5-1.0',
↪  sampling_period=dt, method='fft')

plt.figure(figsize=(7, 5))
plt.pcolormesh(t_pulse, freqs_actual, np.abs(coefs)**2,
↪  shading='gouraud', cmap='hot')
plt.colorbar(label='Energy Density')
plt.xlabel('Time (s)', fontsize=12)
plt.ylabel('Frequency (Hz)', fontsize=12)
plt.title('CWT: Gaussian Pulse (2 Hz band, $U=1.402$)',
↪  fontsize=12)
plt.tight_layout()
plt.savefig('cwt_gaussian_pulse.png', dpi=300, bbox_inches='tight')
print("Saved: cwt_gaussian_pulse.png")
plt.close()

f, t, Zxx = signal.stft(f2, fs=fs, window=win, nperseg=len(win),
                        noverlap=int(0.75*len(win)),
                        ↪  boundary=None, padded=False)

plt.figure(figsize=(7, 5))
plt.pcolormesh(t, f, np.abs(Zxx)**2, shading='gouraud', cmap='hot')
plt.colorbar(label='Energy Density')
plt.xlabel('Time (s)', fontsize=12)
```

```
plt.ylabel('Frequency (Hz)', fontsize=12)
plt.title('STFT: Chirp ($\\alpha=0.50$ s, $U=6.632$)', fontsize=12)
plt.ylim([0, 10])
plt.tight_layout()
plt.savefig('stft_chirp.png', dpi=300, bbox_inches='tight')
print("Saved: stft_chirp.png")
plt.close()

coefs, freqs_actual = pywt.cwt(f2, scales_band, 'cmor1.5-1.0',
↪  sampling_period=dt, method='fft')

plt.figure(figsize=(7, 5))
plt.pcolormesh(t_chirp, freqs_actual, np.abs(coefs)**2,
↪  shading='gouraud', cmap='hot')
plt.colorbar(label='Energy Density')
plt.xlabel('Time (s)', fontsize=12)
plt.ylabel('Frequency (Hz)', fontsize=12)
plt.title('CWT: Chirp (2 Hz band, $U=3.471$)', fontsize=12)
plt.tight_layout()
plt.savefig('cwt_chirp.png', dpi=300, bbox_inches='tight')
print("Saved: cwt_chirp.png")
plt.close()

fig, axes = plt.subplots(3, 1, figsize=(8, 9))
```

```
alphas = [0.10, 0.25, 0.50]
U_values = [22.27, 12.75, 6.632]

for idx, (alpha, U) in enumerate(zip(alphas, U_values)):
    half_len = int(np.ceil(4 * alpha * fs))
    n = np.arange(-half_len, half_len + 1)
    tt = n / fs
    win = np.exp(-(tt**2) / (2 * alpha**2))

    f, t, Zxx = signal.stft(f2, fs=fs, window=win,
    ↪  nperseg=len(win),
                            noverlap=int(0.75*len(win)),
                            ↪  boundary=None, padded=False)

    im = axes[idx].pcolormesh(t, f, np.abs(Zxx)**2,
    ↪  shading='gouraud', cmap='hot')
    axes[idx].set_ylabel('Frequency (Hz)', fontsize=11)
    axes[idx].set_title(f'$\\alpha = {alpha:.2f}$ s, $U =
    ↪  {U:.2f}$', fontsize=12)
    axes[idx].set_ylim([0, 10])
    plt.colorbar(im, ax=axes[idx], label='Energy')

axes[2].set_xlabel('Time (s)', fontsize=11)
plt.tight_layout()
```

```
plt.savefig('parameter_comparison.png', dpi=300,
↪  bbox_inches='tight')
print("Saved: parameter_comparison.png")
plt.close()
```